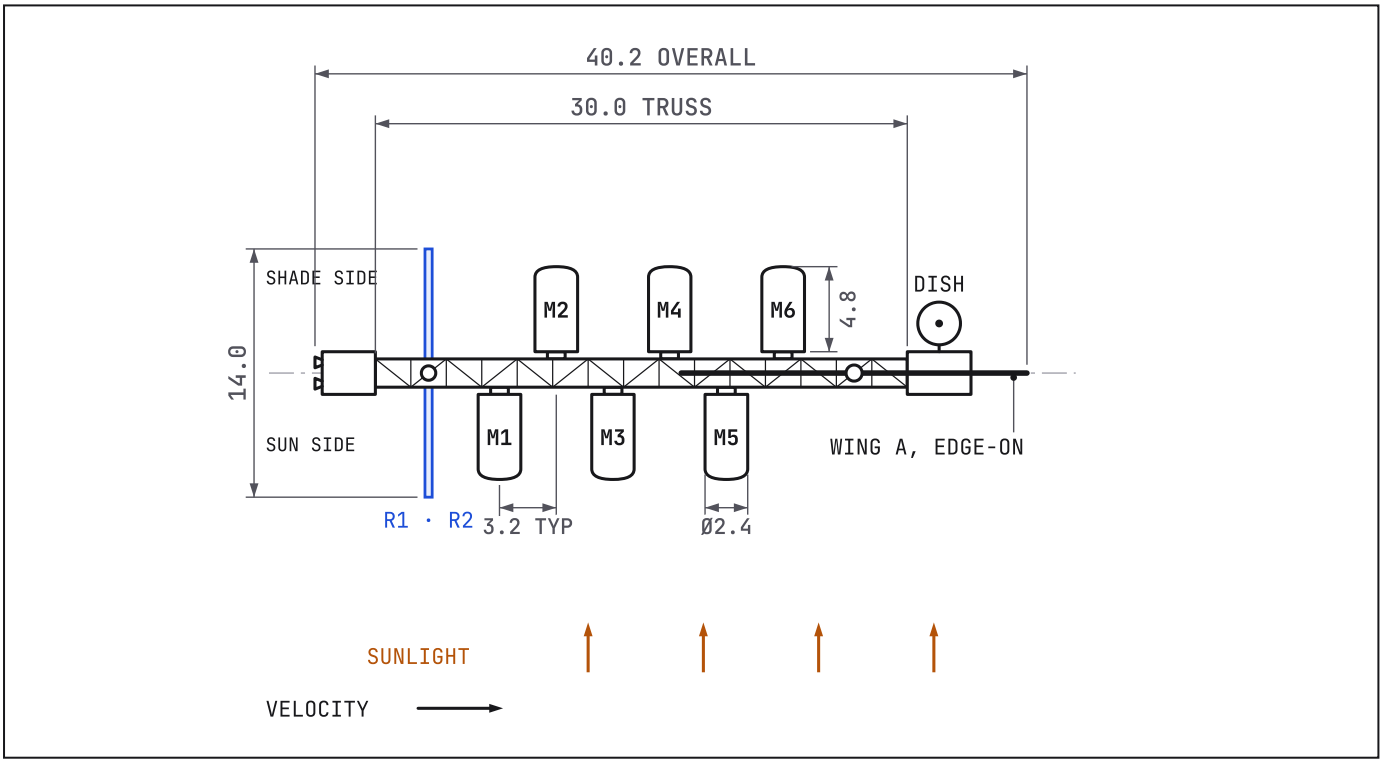
APSIS · KO-CLASS COMPUTE CLUSTER

KO-1 Halyard

General arrangement, as flown in its dawn-dusk orbit

This is how Halyard would be laid out if it had to be built, sized from the same numbers the simulation runs on: 420 kW of compute in six modules, 2,150 m² of radiator, 560 kW of solar. The six modules are the small cylinders on the truss. Nearly everything else on the vehicle exists to feed them power or carry their heat away.

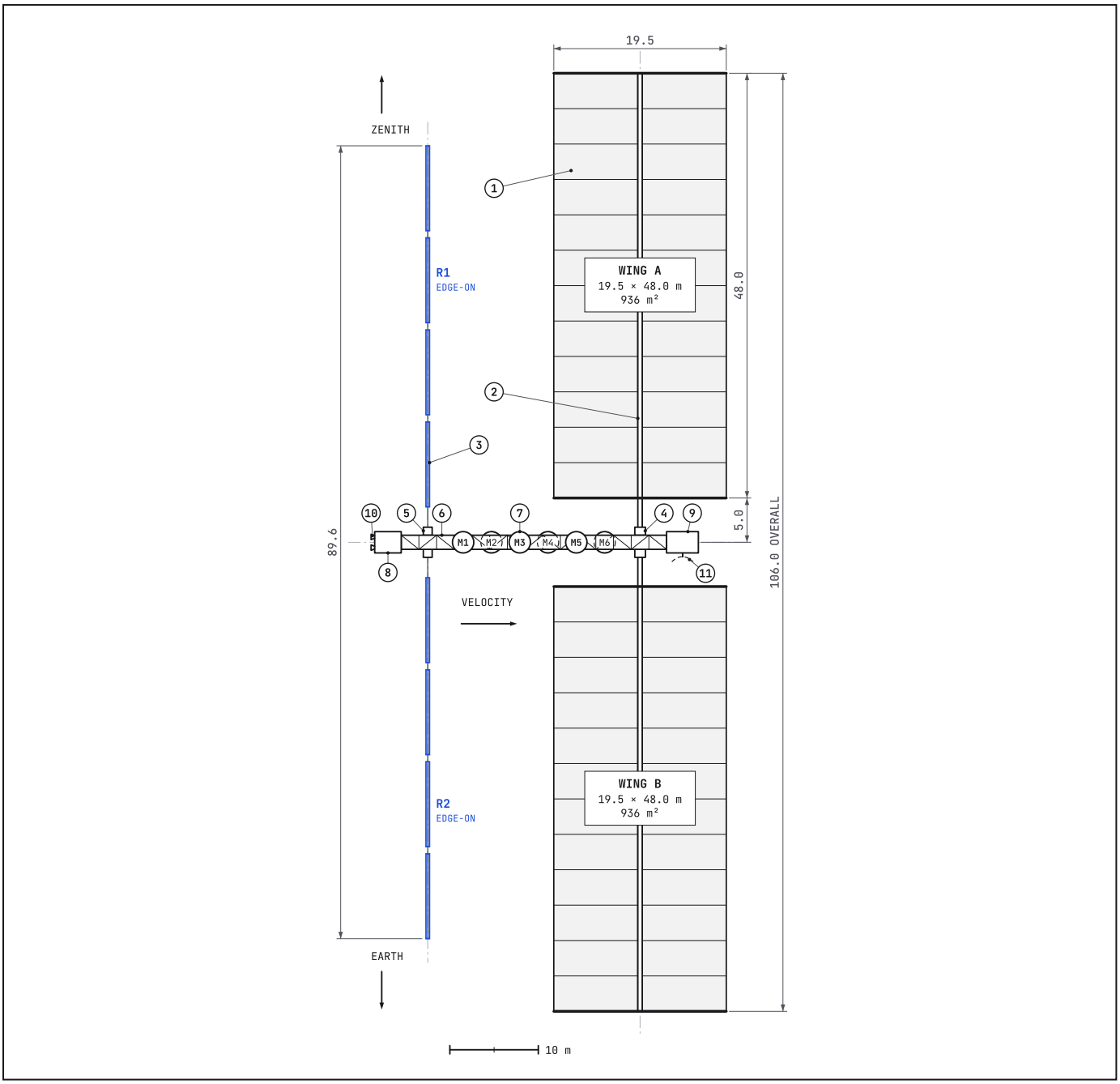
DRAWING APS-KO1-GA-001	REVISION A · concept sizing	UNITS metres	ORBIT SSO dawn-dusk · 550 km
IT LOAD 420 kW	ENVELOPE 106.0 × 40.2 × 14.0	SOLAR ARRAY 1,872 m² · 560 kW	RADIATING AREA 2,150 m²
MASS, ESTIMATED ≈ 40 t			



SHEET 2 OF 5

View from zenith

Looking down the booms. Each module berths at right angles to the truss, so one can be pulled and replaced without disturbing its neighbours. Odd-numbered modules sit on the Sun side.

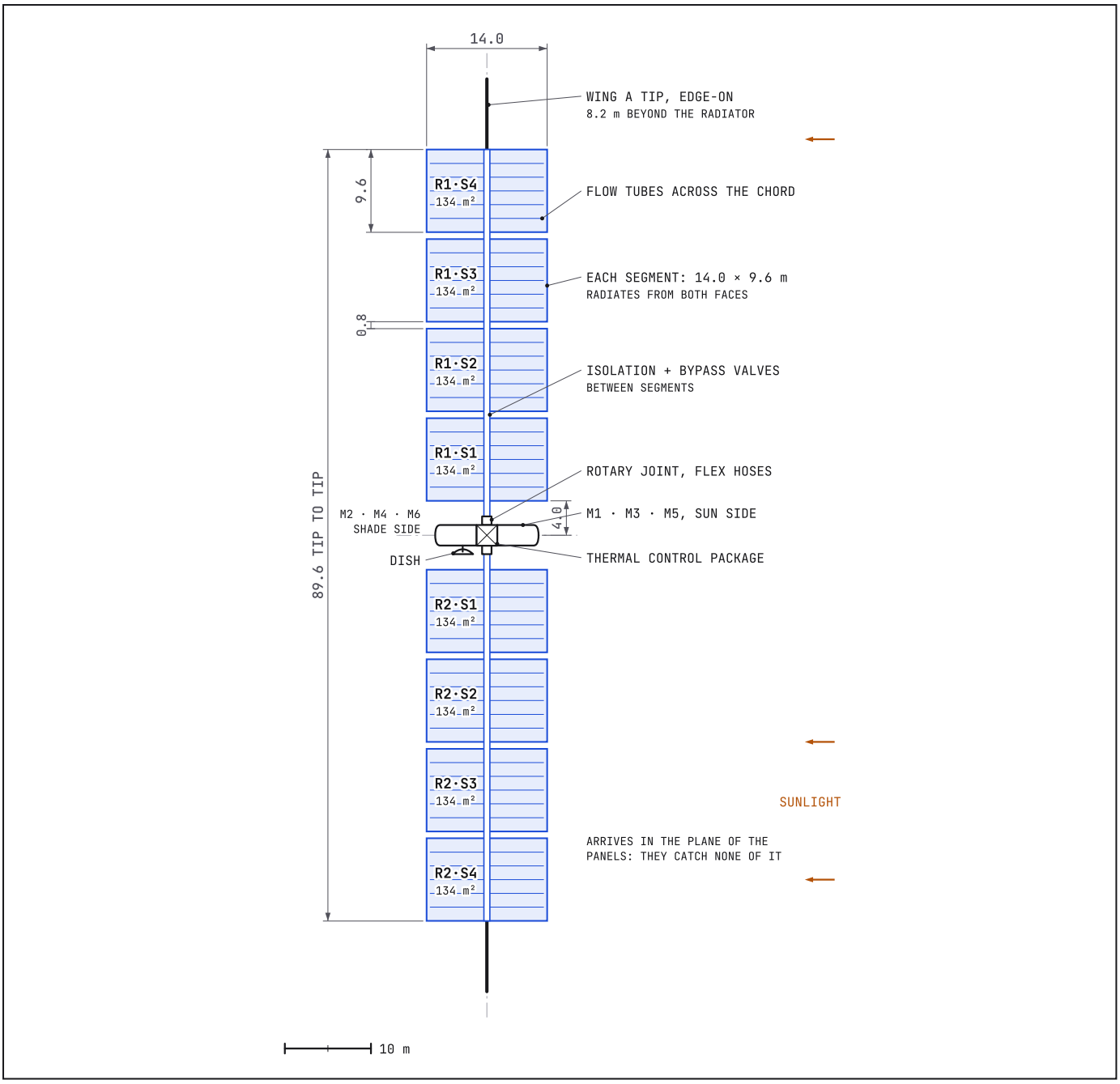


SHEET 1 OF 5

View from the Sun

The vehicle stands 106 m tall with its long axis on the local vertical, which gravity holds steady. The Sun sees both wings square-on and only the thin edges of the radiators.

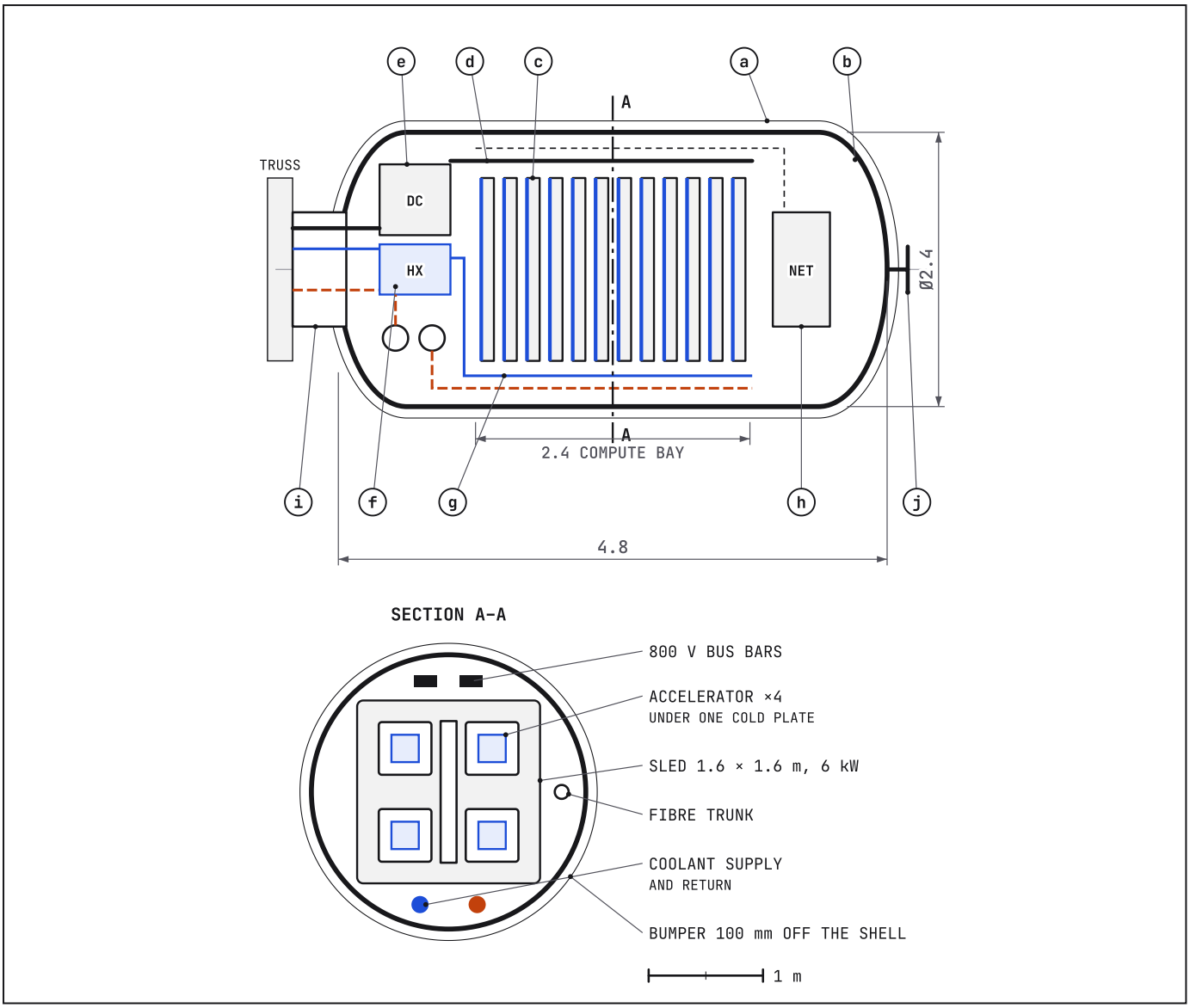
1. Solar blanket, rolled out, two per wing
2. Wing mast, deployable
3. Radiator, seen edge-on
4. Wing drive: turns the wing about its mast to follow the Sun
5. Radiator rotary joint with flexible hoses
6. Truss, 1.6 m square, fifteen 2 m bays
7. Compute module, M1 to M6
8. Thermal control package: pumps P1 and P2, accumulator
9. Bus module: power distribution, 260 kWh battery, avionics, gyros
10. Hall thrusters for drag make-up
11. Ground-link dish on a two-axis gimbal, hidden behind the bus



SHEET 3 OF 5

View from aft

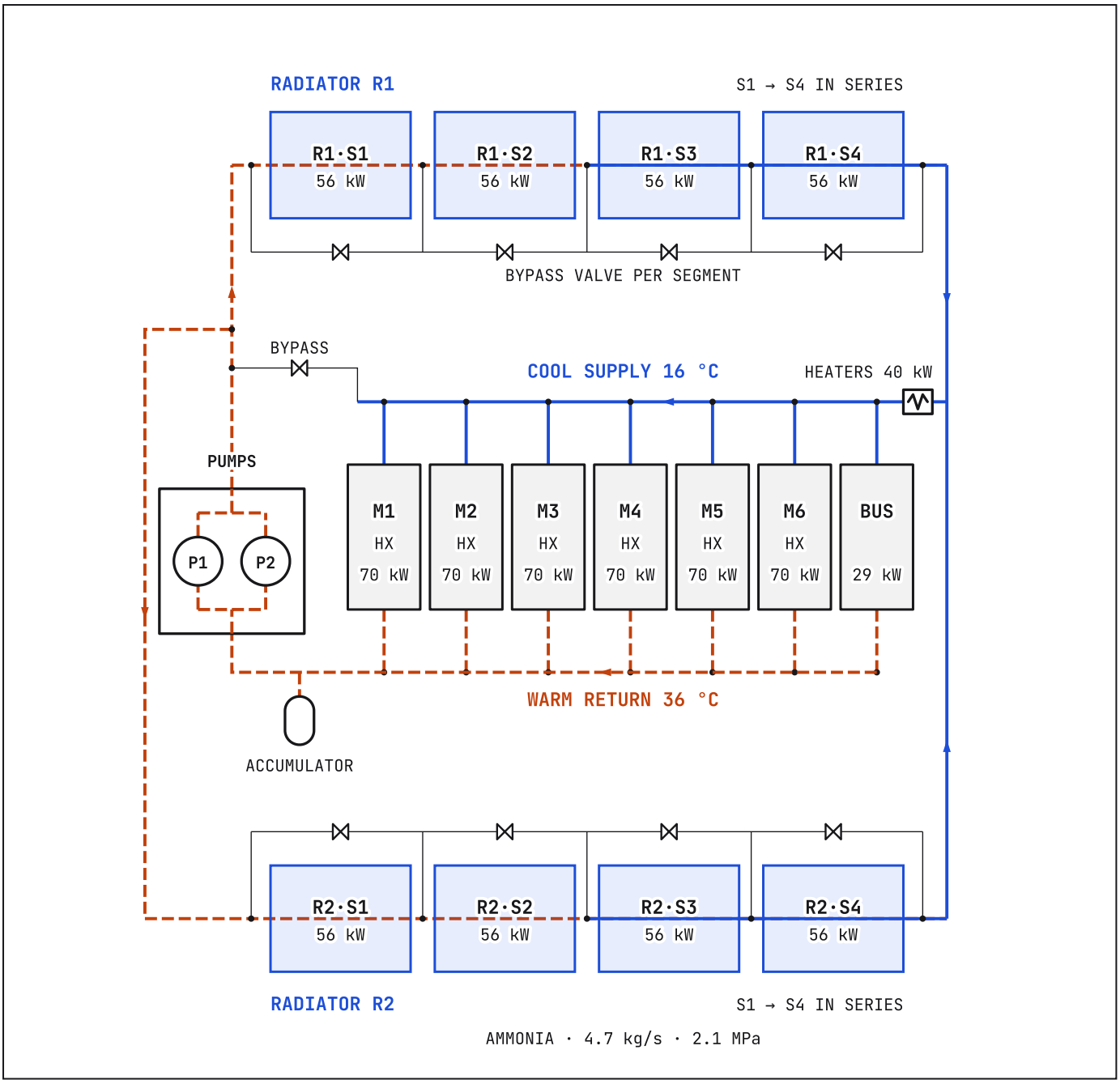
The radiators face fore and aft, toward empty sky. Each of the eight segments is a 14.0 × 9.6 m panel that radiates from both faces. Sunlight travels in the plane of the panels, so they absorb almost none of it.



Compute module, section

One of six. Twelve sleds of four accelerators each, 6 kW a sled, bolted to cold plates. The canister is sealed and filled with dry nitrogen so ordinary server hardware never sees vacuum. About 2.0 t.

- (a) Debris bumper, 1.2 mm aluminium, over an insulation blanket
- (b) Shell, 5 mm aluminium-lithium, dry nitrogen at 50 kPa
- (c) Compute sled, twelve at 0.2 m pitch
- (d) 800 V bus bars
- (e) Power conversion, 800 V to 48 V
- (f) Heat exchanger: the module's water-glycol loop hands heat to the ammonia loop
- (g) Coolant manifolds and two circulation pumps
- (h) Network switch and optical backplane
- (i) Berthing collar: quick-disconnects for ammonia, power and fibre
- (j) Grapple fixture, so a servicer can swap the module



SHEET 5 OF 5

Coolant loop

One pump runs and one is spare. Warm ammonia (dashed) leaves the modules at 36 °C, crosses four radiator segments in series on each side, and comes back at 16 °C (solid). Any segment can be bypassed if it is punctured, at a cost of one eighth of the cooling.

How the big parts were sized

QUANTITY	VALUE	BASIS
HEAT		
Heat to reject	449 kW	420 kW of compute plus 7% for avionics and comms
Radiating area	2,150 m²	8 segments × 134.4 m² × 2 faces
Heat the panels soak up	190 W/m²	Earth's infrared, reflected light and grazing sunlight, as in the simulation
Loop temperature, full load	26 °C	$Q = A(\varepsilon\sigma T^4 - 190)$, with $\varepsilon = 0.88$
Hottest module, full load	43 °C	17 K above the loop; the limit is 45 °C
One segment bypassed	385 kW	92% of compute, holding the loop at 28 °C
Coolant flow	4.7 kg/s	Ammonia warming 20 K across the modules
POWER		
Array output	560 kW	1,872 m² at 299 W/m²: 30% cells, 0.86 packing, 0.85 for heat, pointing and wiring
Load in sunlight	478 kW	Compute 420, avionics 29, pumps 6, conversion losses 23
Margin	82 kW	15%, for cell ageing and battery charging
Battery	260 kWh	The longest eclipse of the year is 22 minutes: 449 kW × 22 min = 162 kWh, 62% of the pack

Mass estimate

ITEM	KG	BASIS
Compute modules, six	12,000	2.0 t each: sleds, cold plates, shell, bumper
Radiators	5,400	1,075 m² of panel at 5 kg/m²
Solar wings	4,700	560 kW at 120 W/kg, masts included
Attitude control and propulsion	2,700	Gyros, Hall thrusters, propellant
Truss and rotary joints	2,500	
Power distribution	2,200	800 V switchgear and harness
Battery	1,750	260 kWh at 150 Wh/kg
Coolant loop	1,600	Pumps, lines, accumulator, about 400 kg of ammonia
Communications	350	Dish and two optical terminals
Margin, 20%	6,640	
Total	**39,840**	**95 kg for each kW of compute**

How the 3D model in the app follows this drawing

- **To scale.** Every cluster's model is built in metres from the dimensions in `src/sim/design.ts`. Halyard is exactly as drawn here. The other clusters share its truss, modules and bus, with wings and radiators sized to their own power and heat at the same proportions.
- **Flight attitude.** A cluster whose orbit keeps the Sun more than 45° out of its orbit plane flies as drawn: booms vertical, Sun broadside. That is the two dawn-dusk clusters and KO-5. The rest fly with their booms across the orbit, where the wings turn once an orbit to follow the Sun.
- **Pipes and cables** are drawn about three times their true bore, so the coolant, data and power can be seen flowing.
- **Inside the modules.** The model shows the canister only. Sheet 4 is the one place the sleds, heat exchanger and power conversion are drawn.
- **Eclipses.** The simulation keeps the Sun fixed, so the dawn-dusk clusters never pass through shadow. At 550 km there is in fact a season around one solstice with up to 22 minutes of shadow an orbit. The battery is sized for that.

This is concept sizing. There is no structural, deployment, launch-packaging or attitude-control analysis behind it, and the mass figures are estimates good to about ±30%.